

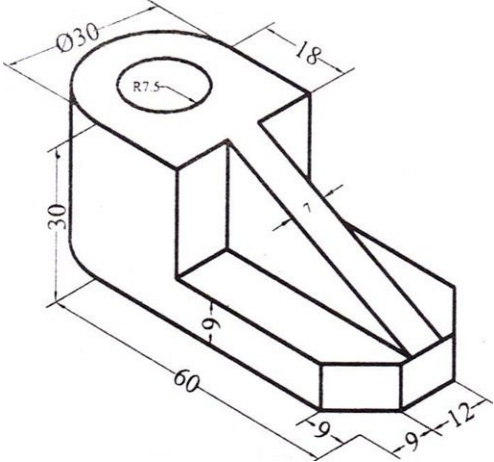
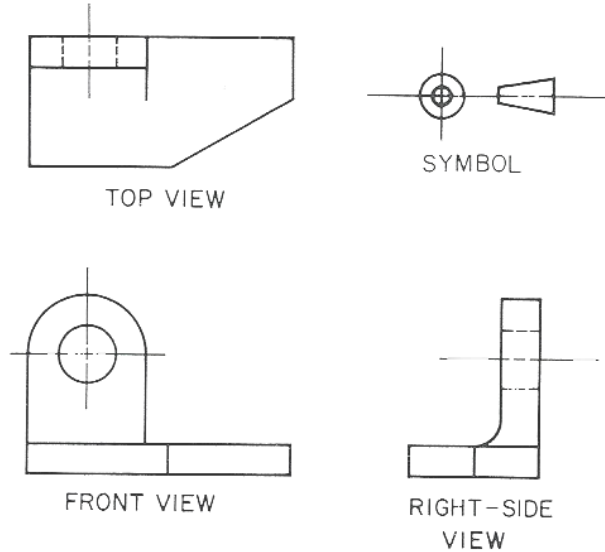
Engineering drawings (Q&A)

Learner: _____ Date: _____

Task 1

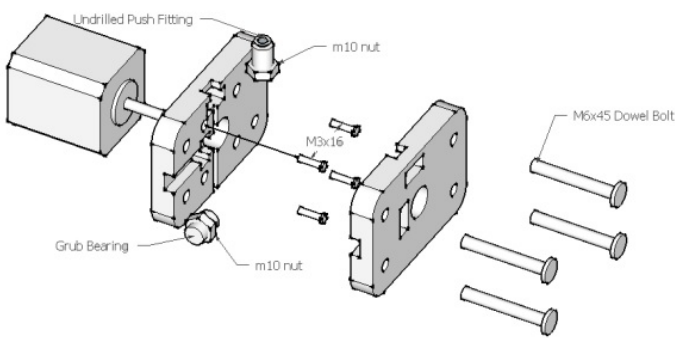
Identify the drawing type and explain what each would be used for.

[assembly drawing – circuit diagram - isometric - third angle projection – block diagram – wiring diagram – general arrangement diagram – fluid power diagram]

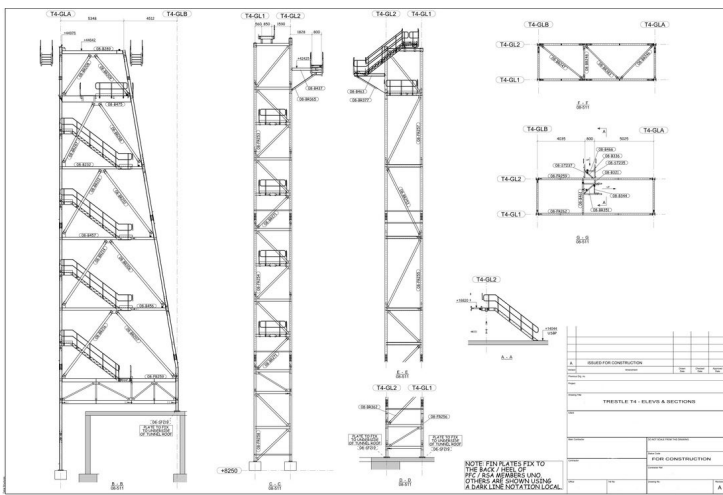
Drawing type	Used for
 Type:	
 Type:	

Drawing type

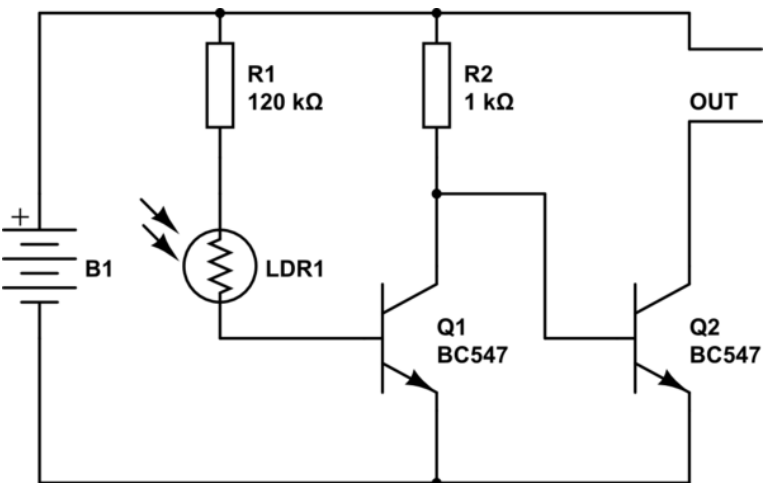
Used for



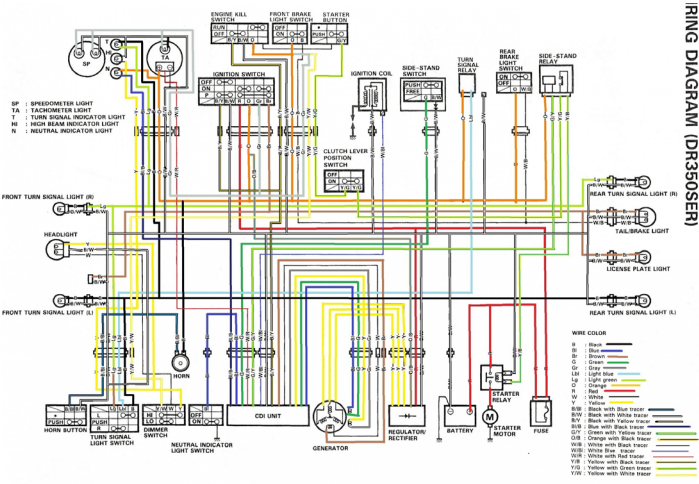
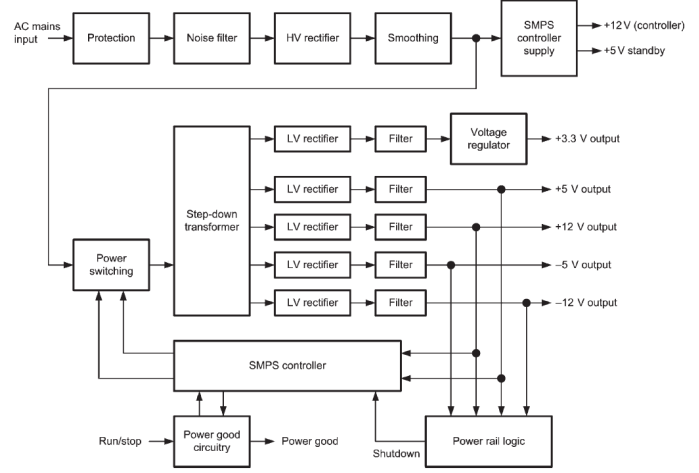
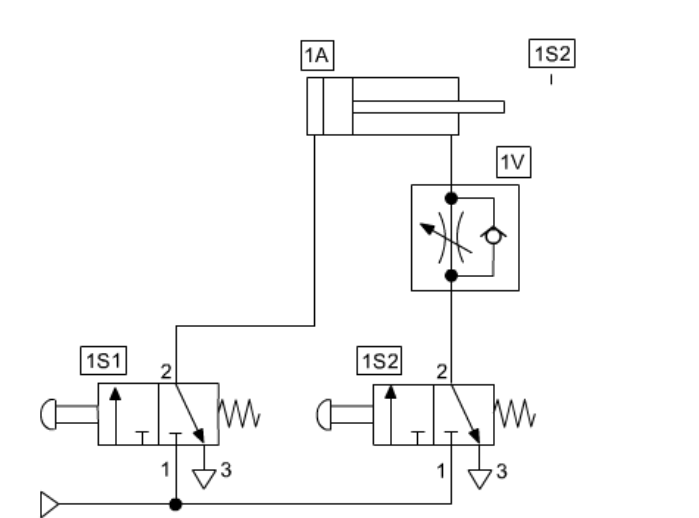
Type:



Type:



Type:

Drawing type	Used for
	
Type: 	
Type: 	

Task 2

For each of the situations given below, state what the additional information would be used for.

Situation 1: Tap/drill chart

Metric Coarse Tapping Drill Sizes

Size	Pitch	Drill
M1	0.25	0.75
M2	0.4	1.6
M3	0.5	2.5
M4	0.7	3.3
M5	0.8	4.2
M6	1	5
M7	1	6
M8	1.25	6.75
M9	1.25	7.75
M10	1.5	8.5
M11	1.5	9.5
M12	1.75	10.2

What would this information be used for?

Situation 2: Bend allowance equation

Sheet Metal Bend Allowance Formula

$$BA = A \frac{\pi}{180} (R + KT)$$

BA – Bend Allowance

A – bend angle in deg

R – inside bend radius in m

K – constant

T – material thickness in m

What would this information be used for?

Task 3

Why might technical information be presented in different ways as shown on the previous pages?

Task 4

Describe each of the following forms of communication and tick the appropriate box for the level of detail they should include.

A: Sketch

☐ basic detail ☐ moderate detail ☐ thorough detail

B: Test/inspection reports

☐ basic detail ☐ moderate detail ☐ thorough detail

C: Work planning document

☐ basic detail ☐ moderate detail ☐ thorough detail

Task 5

Why is it important that sketches are of a suitable size and use appropriate drawing conventions?

Task 6

Why is it important to use a fixed common reference point (datum) when dimensioning a drawing? See the image below as an example

